

ABBA2 Midpoint

Explicit duplicated-space step with arithmetic projection

1 Definition

For $\dot{z} = f(t, z)$, $z \in \mathbb{R}^m$, duplicate the accepted state on the diagonal, $u_0 = v_0 = z_n$, and write $s = h/2$. The endpoint-time ABBA stages are

$$u_1 = u_0 + sf(t_n, v_0), \quad v_1 = v_0 + sf(t_n, u_1), \quad (1)$$

$$v_f = v_1 + sf(t_n + h, u_1), \quad u_f = u_1 + sf(t_n + h, v_f). \quad (2)$$

The physical output is the midpoint of the copies,

$$z_{n+1} = \frac{u_f + v_f}{2}.$$

“Midpoint” therefore refers to this arithmetic projection; the method is not the implicit-midpoint Runge–Kutta rule.

2 State and energy configurations

The model has no multiplier, residual, Newton iteration, or Broyden update. Its state and energy controls are `state_extension` and `track_energy`:

Extension	tracking	accepted state	duplicated base state
<code>physical</code>	off	$z \in \mathbb{R}^{2N}$	$(u, v) \in \mathbb{R}^{4N}$
<code>physical</code>	on	$z \in \mathbb{R}^{2N}$	$(u, v) \in \mathbb{R}^{4N}$
<code>fully_extended</code>	on	$Z = (z, t, k) \in \mathbb{R}^4$	$(Z_1, Z_2) \in \mathbb{R}^8$

Physical tracking reuses the four accepted stage states to advance one $\kappa = k/2$ per particle. This triangular sidecar does not feed back into the physical trajectory and does not change the accepted or duplicated dimensions. Fully extended mode, which currently applies to one guiding-centre particle, always enables energy evolution, duplicates all of Z , and may therefore define a different physical map.

3 Symmetry, order, and geometry

The four shears form an explicit self-adjoint map on the duplicated phase space when the signed step reverses together with its endpoint times. Starting from the diagonal and taking the arithmetic mean yields a consistent symmetric method with designed global order two for sufficiently smooth dynamics.

The duplicated ABBA map is symplectic for its split Hamiltonian. Arithmetic projection, however, is not Hairer’s symmetric projection and does not in general make the closed physical map symplectic. The copy-separation quantity

$$d_n = \|u_f - v_f\|_\infty$$

is therefore retained as a diagnostic. It is neither an error estimator nor a convergence residual.

4 Work and time dependence

One step performs four vector-field evaluations. The evaluations at t_n and $t_n + h$ are part of the non-autonomous definition. The common fixed-grid runner chooses a uniform main grid independently of requested output times; off-grid output uses shadow steps that do not alter the accepted main trajectory or its diagnostics.

5 Implementation correspondence

The public class is implemented in `src/simulation/methods/abba/order2_midpoint.py`. It reuses the common endpoint-time stages from `abba/_core.py`; optional physical energy tracking lives in `abba/_energy.py`, and the fully extended kernel is selected by the same public class. The companion architecture document under `../simulation/` records the complete runtime and observation contract.